Learning Outcomes

1. Apply information and data science concepts and methods by thinking critically and creatively to conceptualize and solve real world problems;
2. Demonstrate an understanding of the data lifecycle, including data curation and stewardship. distributed computing, and the data pipeline ecosystem;
3. Demonstrate an understanding of professional and ethical responsibility in data ownership, security, sensitivity of data, and consequences and privacy concerns of data analysis;
4. Demonstrate an ability to identify user-centric information and data needs of diverse populations and to interpret data findings effectively to diverse audiences, orally, visually, and in written formats;
5. Demonstrate an understanding of how people from diverse backgrounds and cultures interact with technology and with each other in technological contexts; and how to plan strategically for emerging technologies and the changing technological landscape.

Career

Graduates with a BS in Information Science and Data Analytics will have knowledge of software principles and practices, programming concepts and techniques, data structures, and system development methods and practices. In addition, they will have studied the human aspect of technology usage with consideration on privacy and ethics and impacts on society.

With this knowledge and skill set, graduates will be prepared for careers in data analytics, UX/UI, technology development, data management, cybersecurity, web content and development, and information architecture. They may hold positions such as business intelligence analyst, data analytics consultant, front end web developer, data architect, information assurance analyst, and data visualization specialist.

During their time in the program, students can receive career support from the San José State University Career Center.

- **Data Analytics**: Collecting, organizing, transforming, and analyzing data to assist in decision making through predictive analysis.
- **UX/UI**: Applying interactive and visual design concepts to websites and web applications for a positive and cohesive user experience.
- **Software Development**: Creating, designing, deploying, and supporting software development in organizations of all types.

- **Cybersecurity**: Application of technologies, processes and controls to protect systems, networks, devices and data from cyber attacks.
- **Web Development**: Building and maintaining websites.
- **Business Consultant**: Analyze data, processes, and workflows to improve processes, services, and products.
- **Data Analyst**: Make sense of data to empower decision-making through the use of statistical and data interpretation concepts.
- **Data Architect**: Design, implement, and monitor an organization's data management framework to support creation, acquisition, maintenance, and control of enterprise data.
- **Information Assurance Analyst:** Perform assessments against organization policy and regulatory control frameworks to ensure an organization meets customer, regulatory, and industry requirements
- **IT Advisor**: Provide broad technical expertise and assistance related to an organization's IT infrastructure.
- **UI/UX engineer**: Build and design digital products such as websites and apps based on user experience and user interface principles.
- **Data Visualization Specialist**: Create graphs, charts, and multimedia visuals to support comprehension and decision-making

Program Delivery

The 4-year BS in Information Science and Data Analytics program is a hybrid program starting in fall 2024. That means that while most courses will be delivered online in an asynchronous format, some courses will be offered on campus in San Jose. Students applying to the program starting in fall 2024 should anticipate having to take some courses on campus.

All core general education (lower division) courses can be taken in person or online, and can be completed at SJSU or a local community college.

A minor is recommended for students who are so interested. Deciding on a minor should only be finalized after consultation with the major advisor.

Admissions Requirements

Freshmen who meet the CSU eligibility requirements are eligible to apply to the BSISDA program.

Applications for admission are currently only accepted for a start in fall semester. Admissions for fall 2026 will be open from October 1 through December 1, 2025.

Applications to SJSU and the 4-year BSISDA program are handled through the Cal State Apply system. You can find out more about admission to SJSU on the Freshman Steps to Admission page in the SJSU Student Affairs office.

Pre-admissions questions and counseling are available from the SJSU Student Outreach and Recruitment (SOAR) office.

Fees and Scholarships

Fees and Tuition

Information on tuition and fees can be found on the Bursar's Office website.

Students are advised to budget for textbook expenses, as well as obtaining the necessary equipment to prepare their home computing environment, including monthly Internet service, so that they may begin their first semester with all materials in place.

Scholarships

Information on scholarships and aid can be found on the Financial Aid and Scholarships website.

Academics

Unit Breakdown:

Core GE: 39 units

Minor or University Electives and Major Electives: 33 units

SJSU Studies: 12 units

Major Requirements: 36 units

Total: 120 units

A minor is strongly recommended and may be decided on by taking into consideration the career path and a consultation with the advisor.

Required Courses:

ISDA 101 Online Learning Tools and Strategies for Success

All new BSISDA students must complete ISDA 101 within the first four weeks of their first semester. Requests to add the class after the semester has started are not granted without extenuating circumstances and *must be within the first 7 days*. No exceptions.

This **1-unit CR/NC introductory course** covers key technologies and strategies for online learning, including learning management tools, library databases, web conferencing, and more. Delivered 100% asynchronously online.

Any student who does not successfully complete the course in their first semester is placed on academic notice. A registration hold will prevent enrollment in other courses. After a discussion with one of the Program Coordinators, a second attempt may be granted if the student presents compelling reasons for not having completed the course.

Failure to pass ISDA 101 on the second attempt leads to disqualification from the university.

Read more

KIN 169 – Diversity, Stress and Health

Impact of structured inequalities on stress and health of diverse populations. Analysis of physiological/psychosocial health factors related to diversity, as well as behavioral interventions and social actions that mediate stress and optimize health and social justice.

Satisfies SJSU Studies S: Self, Society & Equality in U.S.

Prerequisite(s): Completion of Core General Education and upper division standing are prerequisites to all SJSU studies courses. Completion of, or co-registration in, 100W is strongly recommended.
Grading: Letter Graded

Note(s): Cross-listed with PH 169. Kinesiology is responsible for scheduling.
3 unit(s)

Math Requirement: Elementary Statistics

STAT 95 – Elementary Statistics (3 units)

Hypothesis testing and predictive techniques to facilitate decision-making; organization and classification of data, descriptive and inferential statistics, central tendency, variability, probability and sampling distributions, graphic representation, correlation and regression, chi-square, t-tests, and analysis of variance. Computer use in analysis and interpretation.

Satisfies B4: Mathematics/Quantitative Reasoning.

OR

UNVS 15 – Statway: Statistics-Concepts & Methods (3 units)

Concepts and methods of statistics with an emphasis on data analysis. Topics include methods for collecting data, graphical and numerical descriptive statistics, correlation, simple linear regression, basic concepts of probability, confidence intervals and hypothesis tests for means and proportions, and chi-square tests.

Lecture 1 hour / Activity 4 hours

Satisfies Completion with a grade of C- or better satisfies the GE Area B4: Mathematics/Quantitative Reasoning requirement.

ISDA 20A Query Languages (1 unit)

Introduces students to the features and functionalities of structured query language (SQL) focusing on to manipulating data in a database, creating tables and indexes, modifying stored records, creating database objects.

SJSU students who have completed one of the following courses may use that in place of ISDA 20A:

- CS 157A – Introduction to Database Management Systems
- CMPE 138 – Database Systems
- BUS4 112 – Database Management Systems
- SE 157B – Database Management Systems II
- MATH 167PS – Introduction to Python Programming and SQL

ISDA 20B Python Programming (2 units)

Hands-on introduction to Python focusing on solving a problem by developing flowcharts and algorithms, building a Python application using Python I/O (input/output), variables and data types, conditionals, and loops, and Python packages.

Students who have completed one of the following courses may use that in place of ISDA 20B:

- CS 22A – Python for everyone
- CS 122 – Advanced Programming with Python
- TECH 30 – Introduction to Python Programming
- CS 22B – Python Programming for Data Analysis
- MATH 167PS – Introduction to Python Programming and SQL

ISDA 20C Web Development (1 unit)

Focuses on the structure and functionality of the world wide web and modular web design techniques. Covers HTML5, CSS, and JavaScript. The students will also learn about cultural issues in website designing, and accessible and responsive web design.

Students who have completed the following courses may use that in place of ISDA 20C:

- ART 109 – Web Development in HTML, CSS and JavaScript

ISDA 20D Computer Operating Systems (1 unit)

Overview of the design and implementation of computer operating systems (OS) with emphasis on the three major subsystems: process management (processes, threads, CPU scheduling, synchronization, and deadlock), memory management (segmentation, paging, swapping), and file systems; and distributed systems.

Students who have completed one of the following courses may use that in place of ISDA 20D:

- CS 149 – Operating Systems
- CMPE 142 – Operating Systems Design
- CMPE 180C – Operating Systems Design

LLD 100WB – Writing Workshop

Advanced composition. Instruction and practice in reading, discussing and writing about issues germane to the student's major field of study.

Satisfies SJSU Studies Z: Written Communication II.

Prerequisite(s): A3 or equivalent second semester composition course (with a grade of C- or better); Completion of core GE or completion of ENGL 100A / LLD 100A with C or better; and upper division standing. No Business Administration majors. Or Graduate or Postbaccalaureate level.
Grading: Letter Graded

Note(s): Must be passed with C or better to satisfy the CSU Graduation Writing Assessment Requirement (GWAR).
3 unit(s)

NuFS 139 Hunger & Environmental Nutrition*

Physiology of hunger and malnutrition on human development and health; political, social, cultural, and gender factors that contribute to world hunger; scientific/ technological foundation to population research and food production and their effect on the environment. Meets GE Area R.

Prerequisite(s): Completion of Core General Education and upper division standing are prerequisites to all SJSU studies courses. Completion of, or co-registration in, 100W is strongly recommended.
Grading: Letter Graded

*As an alternative to NuFS 139, students may take any course that meets Area R requirements.

Phil 134 Computers, Ethics and Society*

The nature of privacy in a technologically interconnected world; the role of computer technologies in the exercise of the human intellect and imagination with respect to freedom of expression and the social good; rights and responsibilities of intellectual property ownership.

Satisfies SJSU Studies V: Culture, Civil & Global Understanding.
(3 units)

Prerequisite(s): Completion of Core General Education and upper division standing are prerequisites to all SJSU studies courses. Completion of, or co-registration in, 100W is strongly recommended.
Grading: Letter Graded

*As an alternative to PHIL 134, students may take any course that meets Area V requirements.

ISDA 100 Technical Writing for Information Science Professionals

Introduces students to the principles of effective communication for technical, scientific, and business environments. Students will create written work in a variety of standard formats and use appropriate style to address the needs of targeted audiences. Delivered 100% asynchronously online. (3 units)

Prerequisite: LLD 100WB

ISDA 110 Technology, Culture and Society

Navigating the social, political, legal and ethical issues in data processing and use. Topics include: social and political factors that interact with data mining, visualization and analytics, explainable AI and FATE Delivered 100% asynchronously online. (3 units)

ISDA 111 Information and Data Science

Conceptual foundations and theoretical underpinnings pertaining to the social, political, and economic factors associated with the information society. Definition of information and information science; constructing information individually and collectively; and emerging information and communications technologies. Concepts, techniques, and tools needed to accomplish various facets of data science practice, including data collection and integration, management, analysis, modeling, and visualization. Data science skills such as product creation, communication, evaluation, presentation, and storytelling. Delivered 100% asynchronously online. (3 units).

Prerequisites: ISDA 20A and ISDA 20B (or equivalents)

ISDA 120 Information Security

Topics include compliance and operational security, psychological approaches to social engineering attacks, Web application attacks, penetration testing, data loss prevention, cloud computing security, application programming development security, firewalls and intrusion detection systems. Delivered 100% asynchronously online. (3 units)

Prerequisites: ISDA 20C, ISDA 20D, and ISDA 111

ISDA 121 Human Centered Cybersecurity

Overview of human interactions with cybersecurity technology. Topics: how people react to cybersecurity policies and tools, reasons for behavior, development, and evaluation of tools for assisting people in cybersecurity, human factors involved with social engineering attacks and phishing; ethical hacking. (3 units)

Read more

ISDA 122 Privacy, Security and Ethics

Data ethics and security will be assessed in terms of privacy, data manipulation, data sharing and ownership, and conflict of interest. Through case studies, students will explore real-world situations involving interplay between data security, privacy, and ethics. Delivered 100% asynchronously online. (2 units).

Prerequisite: Either ISDA 120 or ISDA 121

ISDA 123 Information Assurance

Focuses on information assurance frameworks from both technical and social perspectives. Topics covered: confidentiality, integrity, and availability, security policies, authentication, data protection and access control, risk management, threat and vulnerability assessment, common attack/defense methods, ethical issues. Delivered 100% asynchronously online. (2 units).

Prerequisite: ISDA 122

ISDA 130 User Centered Interface Architecture and Prototyping

Focuses on ways to involve the user at different stages of interface design for web- and mobile- environments. Students will learn, apply, and evaluate various user interface

architectures, modes of interaction, and prototyping, testing, and evaluation techniques. (3 units)

Read more

ISDA 131 Information Architecture

Project-based course developing user-centered knowledge structures for Web environments. Core topics are problems addressed by effective IA; designing for findability; user research; best practices; methods for organizing, labeling, and structuring navigation systems; creating client documents and deliverables. (3 units)

Read more

ISDA 140 Big Data Analytics and Management

Big data technologies and trends including large-scale databases, map-reduce paradigm, big data mining, and big data platforms. Focus is on hands-on learning with tools such as Splunk and Scala/Hadoop using as applied to real-world data sets.

Read more

ISDA 141 Information Visualization

Background of information visualization; perceptual and design principles of information visualization; data analysis methods and hands-on applications of visualization techniques as applied to structured and unstructured data; interaction and interface design issues; social visualization and visual analytics. (3 units)

Read more

ISDA 170 The Emerging Future: Technology Issues and Trends Capstone

Students will gain skills to plan strategically for the changing technological landscape. The course design enables students to participate in futuring activities through readings, video presentations, guest lectures, introductions to today's futurists, and individual research. After a historical overview of forecasting students will become immersed in the literature and practices of current technological futurists and emerging technologies, gain skills to plan strategically for the changing technological landscape, develop a grant proposal outline to implement a new technology, and create a multi-media presentation or infographic that focuses on a new technological area that can impact the future. Possible topics: Blockchain, AR/VR, Voice-based Assistants. Delivered 100% asynchronously online. (4 units)

Prerequisite: Must be taken in last semester of ISDA program

Minor

A minor is strongly recommended and may be decided by taking into consideration the chosen career path and consultation with the advisor.

Some suggested minors are as follows:

- Interaction Design minor – Design Department
- Human Systems Integration minor – Psychology Department
- Values, Technology and Society minor – Anthropology Department
- Digital Media Art minor – Art and Art History Department
- Digital Media for Mass Communications minor – School of Journalism
- Geographic Information Systems minor – Urban and Regional Planning Department
- Computer Science minor – Computer Science Department
- Business minor – College of Business

Elective options open to ISDA majors

Major electives may be selected from courses in the ISDA program such as ISDA 150 – Special Topics in Information Science and Data Analytics, ISDA 160 – User Experience (UX) Research and Design, OR

- Upper division courses in other departments (courses numbered 100-199), as well as
- Approved non-duplicative graduate courses in the School of Information in special topics areas such as
    - INFO 246 – Information Technology Tools and Applications,
    - INFO 287 – Seminar in Information Science, or
    - INFM courses in data analytics, data security, and related topics.

Note: If electives from graduate courses in the School of Information are used to meet the elective requirements, the courses can only be used for the bachelor's degree and cannot be later applied to any graduate degree.

In addition, tuition for elective courses outside the ISDA program may be different than tuition in the program.

Advising

The university's Student Outreach and Recruitment office is available on live chat or via Zoom to assist with questions about San José State University and its programs, as well as the admissions process.

Advising appointments are available virtually and must be scheduled via Spartan Connect.

Students can also contact Kelly Rausch, BSISDA 4-year program coordinator, with any content questions related to course selection and career path.

University Academic Notice and Disqualification

Requirement

All undergraduate students are required to maintain an overall GPA of 2.0 or higher in order to remain in good academic standing and to graduate.

The overall GPA is different from a semester GPA. The overall GPA is your cumulative GPA that you earn starting from the first semester as an undergraduate student at

SJSU. It is the GPA that takes into account the GPA earned for all previous semesters. The semester GPA is your GPA that is calculated on a semester-by-semester basis.

University Academic Notice and At-Risk

An undergraduate student is on Academic Notice if the SJSU GPA falls below 2.0.

If an undergraduate student is currently in good standing, and at the end of a semester their overall GPA falls below 2.0, they will be placed on “University Academic Notice”. A status of University Academic Notice indicates that the student is not performing to minimum undergraduate GPA standards and is at risk of being disqualified from the university.

Once on University Academic Notice, undergraduate students are subject to disqualification if they fail to earn a SJSU term GPA of 2.0 or better each semester until a SJSU cumulative GPA of 2.0 is again established. See: www.sjsu.edu/senate/S16-16E.pdf

Freshmen who meet the CSU eligibility requirements may apply to the BSISDA program.